

How and when is boundary-spanning behaviour related to innovative behaviour: role of intrinsic motivation and organisational support

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Abstract

Purpose – This study aimed to investigate the underlying boundary conditions under which boundary-spanning behaviour has a positive or negative effect on innovative behaviour.

Design/methodology/approach – A multi-wave and multi-source research design was adopted to collect data. Data were analysed using the multilevel structural equation modelling and latent moderated structural equation approach.

Findings – The results showed that boundary-spanning behaviour was significantly and negatively associated with employees' innovative behaviour via ego depletion when employees' intrinsic motivation or organisational support was low. Additionally, boundary-spanning behaviour was significantly and positively associated with employees' innovative behaviour via ego depletion when employees' intrinsic motivation or organisational support was high.

Originality/value – This research suggests that the consequences of boundary-spanning behaviour are conditional, explaining the contrasting conclusions in this regard.

Keywords Boundary-spanning behaviour, Innovative behaviour, Ego depletion, Intrinsic motivation, Organizational support

Paper type Research paper

1. Introduction

Boundary-spanning behaviour, defined as individuals gathering and understanding relevant external information from outside the department/team or the organisation and transmitting this information to team members (Katz and Tushman, 1983), has attracted considerable interest in recent years because it benefits organisations and employees in many aspects (Marrone *et al.*, 2007; Samer and Aimin, 2009; Marrone, 2010; Liu *et al.*, 2018). For example, previous studies have shown that boundary-spanning behaviour is positively associated with team performance, team innovation, team viability, employee promotion, employees' task performance and enhanced reputation (Katz and Tushman, 1983; Marrone *et al.*, 2007; Faraj and Yan, 2009; Marrone, 2010; Lin and Li, 2013; Liu *et al.*, 2018).

However, little is known about boundary-spanning behaviour. Furthermore, our theories and research fail to explain how employees react to their experiences of engaging in boundary-spanning behaviour, which is vital because these experiences have different impacts on employees. On the one hand, as scholars have frequently advanced, boundary-spanning behaviour has positive impacts on employees (Katz and Tushman, 1983; Liu *et al.*,



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2018). On the other hand, the opposite is also true: engaging in boundary-spanning behaviour may drain and deplete actors (Marrone *et al.*, 2007). Indeed, boundary-spanning behaviours require considerable investment in cognitive, emotional and physical resources (Xu, 2019), resulting in feelings of fatigue.

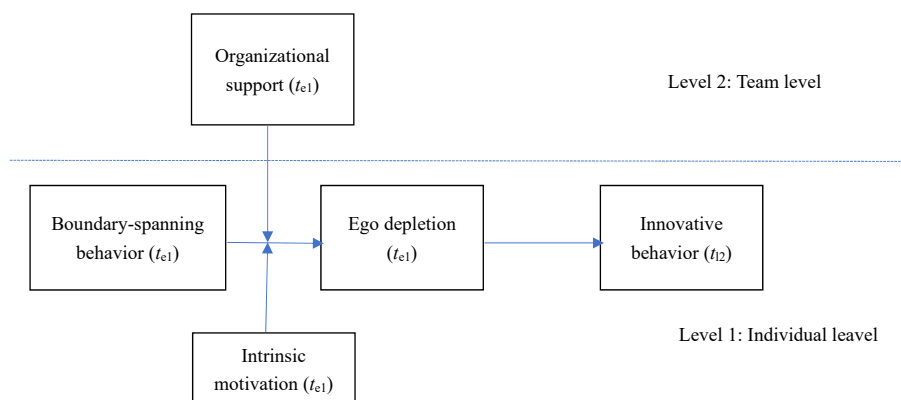
Compared with other constructive behaviours, such as voice, proactive behaviour and ethical behaviour, boundary spanners are expected to gain valuable information from huge, complex and rapidly changing external situations (Katz and Tushman, 1983) and face simultaneous and often conflicting pressures (Marrone *et al.*, 2007), which will consume serious boundary spanners' resources and lead to ego depletion (Baumeister and Vohs, 2007). Ultimately, boundary-spanning individuals may react negatively to future volitional actions. This possibility is supported by Bolino *et al.* (2015) and Bolino and Klotz (2015), who found that engaging in organisational citizenship behaviour would lead actors to experience fatigue, role overload and job stress. The lack of attention paid to the negative consequences of boundary-spanning behaviour for actors makes it unclear how and when boundary-spanning behaviour negatively impacts their subsequent behaviours.

In this study, relying on the ego depletion theory from the perspective of resources, we propose that boundary-spanning behaviour is likely to consume employees' resources and then be associated with ego depletion, a type of short-term mental fatigue (Inzlicht *et al.*, 2014). Ego depletion refers to a decrease in the ability or willingness to carry out subsequent volitional actions (such as controlling oneself, making choices and initiating actions) because of diminished self-control resources caused by performing prior volitional actions (Baumeister *et al.*, 1998).

What do employees do after facing ego depletion? It has been suggested that, after experiencing ego depletion, employees are less likely to engage in subsequent volitional actions, especially those of complex thinking and generating novel ideas (Schmeichel *et al.*, 2003), which are key elements of innovative behaviour, defined as the generation and implementation of novel ideas (Scott and Bruce, 1994). Thus, ego depletion is likely negatively associated with employees' innovative behaviour.

More importantly, based on the ego depletion theory, the association between boundary-spanning behaviour and ego depletion is likely to be influenced by intrinsic motivation (an essential individual's internal resource) and organisational support (a vital external resource) (Baumeister *et al.*, 1998; Inzlicht and Schmeichel, 2012). Intrinsic motivation, which refers to the desire to spend effort based on interest in and enjoyment of the work itself (Amabile, 1993; Grant, 2008), can lead employees to allocate more resources to boundary-spanning behaviour (Inzlicht and Schmeichel, 2012). Organisational support, referring to employees' beliefs concerning the extent to which the organisation values their contributions and cares about their well-being (Eisenberger *et al.*, 1986, 2001), can provide employees with additional resources when needed, helping them to reserve their resources (Baumeister *et al.*, 1998; Siegel, 2011). These will likely act as buffers against the ego-depletion effect (Siegel, 2011). However, previous research has not sufficiently integrated intrinsic motivation and organisational support into boundary-spanning behaviour research. Studying their effects seems essential because it will be helpful to identify when boundary-spanning behaviour may harm the actors themselves (see Figure 1 for our theoretical model).

Our study makes three theoretical contributions to the literature. First, we provide a refined understanding of boundary-spanning behaviour and its effect by examining how and when boundary-spanning behaviour may have a detrimental effect on innovative behaviour. Second, we enrich the literature on the antecedents of innovative behaviour. This study considers individual behaviour and explores how employees' previous behaviour impacts their subsequent innovative behaviour. Third, we complement the theoretical framework of ego depletion theory and provide empirical evidence by testing intrinsic motivation and organisational support as boundary conditions of the ego depletion effect.



Note(s): t_{e1} stands for variables rated by employees at time 1. t_{12} stands for variables rated by leaders at time 2

Source(s): Author's own creation/work

Figure 1.
Theoretical model

2. Theory and hypotheses

2.1 Ego depletion theory

The ego depletion theory stems from research on self-control and proposes that performing volitional actions consumes the limited self-control resources of individuals and then reduces the amount of self-control resources available for subsequent volitional actions (Muraven and Baumeister, 2000). Many empirical studies in different social sciences supported the ego depletion effect in the nineties and early 2000s (e.g. Baumeister *et al.*, 1998; Muraven *et al.*, 1998; Martijn *et al.*, 2007; Wright *et al.*, 2003). However, several recent studies have questioned the existence of an ego-depletion effect (e.g. Hagger *et al.*, 2010; Lin *et al.*, 2020). Notably, a few studies reported non-significant ego-depletion effects (Stillman *et al.*, 2009; Wright *et al.*, 2008). These inconsistent findings illustrate that the ego depletion effect is not unequivocal and that some moderators may influence the ego depletion effect.

Baumeister and Vohs (2016) complemented the ego depletion theory and argued that motivation, beliefs and other situational factors can effectively counteract ego depletion. They suggested that the ego depletion effect no longer indicated exhaustion of resources, but rather unsustainable consumption of resources over a long period, thus leading to the conservation of self-control resources as much as possible. Therefore, individuals invest fewer resources in subsequent volitional actions to reduce further consumption of self-control resources. However, the ego depletion effect does not occur if people perceive that they still have sufficient self-control resources.

The ego depletion theory suggests that motivation moderates the process of ego depletion (Baumeister *et al.*, 2007; Inzlicht and Schmeichel, 2012). When motivated, individuals allocate sufficient resources to carry on volitional behaviours (Inzlicht and Schmeichel, 2012) and perceive that they possess adequate resources. Therefore, individuals with high motivation will not feel fatigued despite the consumption of resources through volitional behaviours. Simply put, the ego-depletion effect does not occur.

Moreover, organisational support also moderates the process of ego depletion (Baumeister *et al.*, 1998; Hobfoll, 2001; Siegel, 2011) because it can provide individuals with an additional provision of resources, saving the individual's own resources, mitigating individual resource loss and resulting in low levels of ego depletion (Siegel, 2011). In contrast, if the organisation does not support or even hinders individuals from performing volitional behaviours, there will be more individual resource consumption, resulting in a higher level of ego depletion.

2.2 Boundary-spanning behaviour and innovative behaviour

Existing research on boundary-spanning behaviour has primarily focussed on its benefits from the perspective of social learning and social networks (Fleming and Waguespack, 2007; Liu *et al.*, 2018; Salem *et al.*, 2018). For example, Salem *et al.* (2018) found that boundary-spanning behaviour could increase learning in humanitarian aid field offices. Song *et al.* (2017) found that boundary-spanning behaviour could increase team innovation via social learning. Additionally, Liu *et al.* (2018) found that boundary-spanning behaviour could increase employees' task performance via social networks.

However, several scholars have recently proposed that boundary-spanning behaviour is likely to be detrimental to boundary spanners by consuming time and effort (Marrone *et al.*, 2007; Song *et al.*, 2017), perhaps implying the adverse effects of boundary-spanning behaviour from the perspective of resources. Meanwhile, sufficient time and resources stimulate innovative behaviour (Amabile, 1993). If resources are insufficient, individuals will be unable to recognise problems or generate novel ideas for solutions (De Jonge *et al.*, 2012). Taken together, we argue that boundary-spanning behaviour is likely to consume individual resources and impact individual innovative behaviour. Therefore, from the context of resources, we integrate the ego depletion theories to identify how and when boundary-spanning behaviour impacts innovative behaviour.

2.3 Mediating effect of ego depletion

The ego depletion theory suggests that performing volitional actions consumes limited individual resources and leads to ego depletion, adversely affecting subsequent volitional actions (Baumeister *et al.*, 1998). In other words, ego depletion links previous behaviours with subsequent ones (Baumeister *et al.*, 2007). Boundary spanners are expected to pay sustained attention to gain valuable information for the team/organisation from complex and rapidly changing external situations and transfer the information to the team/organisation (Katz and Tushman, 1983). This requires boundary spanners to continually focus on recognising valuable information and heavily consume their self-control resources, which is further associated with ego depletion. Moreover, Schmeichel *et al.* (2003) revealed that attention regulation could deplete self-control resources and lead to ego depletion. Additionally, boundary spanners must synthesise information into knowledge. This requires complex higher-order cognitive processing, which consumes tremendous self-control resources (Schmeichel *et al.*, 2003).

Boundary spanners need to play different roles when connected with various outsiders, and there are varied and conflicting requirements during different role-plays (Ramarajan *et al.*, 2011). Thus, boundary spanners must balance these roles and solve the dilemma of different, even conflicting, requirements, which can consume self-control resources and lead to ego depletion (Baumeister and Vohs, 2007).

As individual self-control resources are limited (Baumeister *et al.*, 1998), boundary spanners conserve their resources after resource-consuming boundary-spanning behaviour, leading to insufficient resources to perform innovative behaviour. However, sufficient resources are necessary to perform innovative behaviour (Amabile, 1993). Employees need to focus on the tasks and generate novel ideas in the process of innovative behaviour (Scott and Bruce, 1994), which requires complex higher-order cognitive processing and self-control resources (Schmeichel *et al.*, 2003). Schmeichel *et al.* (2003) also revealed that ego depletion impaired high-level cognitive processes; therefore, ego depletion impairs innovative behaviour. Moreover, Price and Yates (2015) showed that ego depletion could reduce students' creativity. Thus, we propose the following hypothesis:

- H1. Boundary-spanning behaviour has a negative effect on boundary spanners' subsequent innovative behaviour via ego depletion.

2.4 Moderating effect of intrinsic motivation

As we have pointed out earlier, the ego depletion theory suggests that motivation is likely to moderate the process of ego depletion (Baumeister *et al.*, 2007; Inzlicht and Schmeichel, 2012), as motivation influences individual resource allocation (Inzlicht and Schmeichel, 2012). Intrinsic motivation is defined as the desire to expend effort based on interest and enjoyment (Amabile, 1993; Grant, 2008). Following the definition of intrinsic motivation (Amabile, 1993; Grant, 2008), we define intrinsic motivation towards boundary-spanning behaviour as the desire to expend effort based on interest in and enjoyment of boundary-spanning behaviour itself. Boundary spanners with high intrinsic motivation are more likely to allocate more resources to boundary-spanning behaviour because they like doing so. This allows boundary spanners to have sufficient resources to perform boundary-spanning behaviour, limiting ego depletion. In addition, those with more resources are less vulnerable to resource loss (Hobfoll, 2001). Boundary spanners with high intrinsic motivation tend to believe that their resources are unlimited, thus mitigating ego depletion (Job *et al.*, 2010).

In contrast, boundary spanners with low intrinsic motivation are more likely to allocate fewer resources to boundary spanning. This results in employees having insufficient resources to engage in boundary-spanning behaviour, producing ego depletion as soon as they perform boundary-spanning because those with fewer resources are more vulnerable to resource loss (Hobfoll, 2001). Additionally, boundary spanners with low intrinsic motivation do not like such behaviours, but they nonetheless have to do them. Thus, they will have to consume many resources to deal with the imbalance, thereby increasing ego depletion. This argument is supported by Vohs *et al.* (2008), who found that only those who disliked a decision-oriented task felt depleted. Those who enjoyed it showed low depletion effects. Hence, the following hypotheses are proposed:

H2a. Intrinsic motivation moderates the positive relationship between boundary-spanning behaviour and ego depletion, such that the relationship is positive for boundary spanners with low intrinsic motivation and negative for boundary spanners with high intrinsic motivation.

H2b. Intrinsic motivation moderates the mediated relationship between boundary-spanning behaviour and innovative behaviour via ego depletion, such that the mediating effect is negative for boundary spanners with low intrinsic motivation and positive for boundary spanners with high intrinsic motivation.

2.5 Moderating effect of organisational support

The ego depletion theory and related literature suggest that organisational support acts as a buffer against resource depletion (Baumeister *et al.*, 1998; Hobfoll, 2001; Siegel, 2011) because it helps enlarge the available resource pool (Hobfoll *et al.*, 1990). George *et al.* (1993) suggested that organisational support is indicative that help and resources are available from the organisation when needed, that is, to help the employee to continue to perform well in the case of high work demands. Therefore, boundary spanners with high organisational support are more likely to be provided with more encouragement and assistance by their organisations when they engage in boundary-spanning behaviour. Thus, they are more likely to rely on organisational resources to perform boundary-spanning behaviour, which could mitigate resource loss and result in a low level of ego depletion (Siegel, 2011). More importantly, high organisational support makes boundary spanners perceive that the organisation values their contributions and cares about their well-being (Eisenberger *et al.*, 2001), which encourages their critical psychological resources (Hobfoll *et al.*, 1990). This widens the available resource pool of boundary spanners, thereby limiting ego depletion. On the contrary, if the organisation does not support boundary spanners and lets them work alone, they will have to consume many resources of their own and are more likely to feel ego depletion. Hence, the following hypotheses are proposed:

- H3a.* Organisational support moderates the positive relationship between boundary-spanning behaviour and ego depletion such that the relationship is positive for boundary spanners with low organisational support and negative for boundary spanners with high organisational support.
- H3b.* Organisational support moderates the mediated relationship between boundary-spanning behaviour and innovative behaviour via ego depletion, such that the mediating effect is negative for boundary spanners with low organisational support and positive for boundary spanners with high organisational support.

3. Method

3.1 Participants and procedure

Participants were recruited from 38 companies, including banks, consulting companies, real estate companies and Internet companies located in Beijing, Shenzhen and Shanghai, China. Multi-wave and multi-source approaches were used. With the help of close friends, we got in touch with the human resource managers of each company, who helped us mobilise their employees and leaders to participate in our survey, and subsequently, we got the roster of voluntary participants. Each employee was paid 20 Chinese yuan (approximately 3.13 USD), and each leader was paid 50 Chinese yuan (approximately 7.83 USD). Each employee was numbered using an identification code that allowed them to match their leaders. At Time 1, paper-and-pencil surveys were distributed to 620 employees during their breaks at work. Among them, 597 employees returned the surveys to their human resource managers, who then sent them to us. Employees rated boundary-spanning behaviour, intrinsic motivation, ego depletion, organisational support and demographic information. Approximately one month later, 135 leaders rated these employees' innovative behaviour and team type and reported their demographic information. A total of 120 leaders returned the survey. We were able to pair 519 employees (83.71%) with their leaders to obtain 108 (80.00%) matching data. The number of employees in each team ranged from three to seven. *T*-tests and chi-square tests were conducted to examine the impact of attrition, and the results of *t*-tests showed no significant differences in employees' age ($M_{\text{nonresponders}} = 30.36$, $M_{\text{responders}} = 29.87$, $t = 0.76$, $p > 0.05$) and boundary-spanning behaviour ($M_{\text{nonresponders}} = 4.87$, $M_{\text{responders}} = 4.79$, $t = 0.45$, $p > 0.05$) between the two groups of respondents and non-respondents (Dooley and Lindner, 2003). The results of the chi-square test showed no significant differences in employees' marital status ($\chi^2 = 3.08$, $p > 0.05$) but significant differences in employees' gender ($\chi^2 = 13.35$, $p < 0.05$) between the two groups of respondents and non-respondents. Thus, we controlled for gender to reduce attrition bias when testing our theoretical model (Collins *et al.*, 2001).

Among the employees, 55.50% were women, 54.50% were married and more than half (62.80%) had earned a college degree or higher. Their average age and tenure respectively were 29.87 years old ($SD = 4.34$) and 3.46 years old ($SD = 3.50$), respectively. Among the leaders, 45.40% were women, 79.60% were married and more than half (67.60%) had earned a college degree or higher. Their average age and tenure respectively were 34.94 years old ($SD = 4.58$) and 5.76 years old ($SD = 3.89$), respectively.

3.2 Measures

3.2.1 Boundary-spanning behaviour. Faraj and Yan's (2009) four-item scale was adopted to measure employees' boundary-spanning behaviour (Cronbach's $\alpha = 0.79$). A sample item was "I solicit information and resources from elsewhere in and/or beyond the division" (from 1 = strongly disagree to 7 = strongly agree).

3.2.2 Intrinsic motivation towards boundary-spanning behaviour. After measuring employees' boundary-spanning behaviour, we asked the employees, "Why do you engage in these behaviours?" (from 1 = strongly disagree to 5 = strongly agree). Four items from Grant's (2008) scale were adopted (Cronbach's alpha = 0.96). A sample item was "Because I enjoy it".

3.2.3 Ego depletion. We employed the five-item scale (Cronbach's alpha = 0.97) of Johnson *et al.* (2014) to measure employees' ego depletion. We asked the employees to rate how they felt after having performed boundary-spanning behaviour. A sample item was "I feel mentally exhausted", (from 1 = strongly disagree to 5 = strongly agree).

3.2.4 Organisational support. The six-item scale of Eisenberger *et al.* (2001) was used to measure organisational support. A sample item was "The organisation is willing to help me if I need a special favour", (from 1 = strongly disagree to 7 = strongly agree). One of the six items is reverse-scoring item ("The organisation shows little concern for me (R)"). The standardised coefficients of factor loading for the reverse-scoring item was less than 0.5, thus we dropped the reverse-scoring item [1]. Cronbach's alpha for five other items was 0.82. We then aggregated individual responses to the team level by calculating the mean of organisational support by all team members within each team. We computed r_{wg} , ICC(1) and ICC(2) to justify aggregation (James, 1982; Lebreton and Senter, 2007). The results showed median r_{wg} is 0.94 (0.37–1, mean = 0.91, SD = 0.09), ICC(1) is 0.25, whereas ICC(2) is 0.61. Overall, these results support the aggregation, although higher ICC(2) values would have been preferable.

3.2.5 Innovative behaviour. Leaders rated employees' innovative behaviour using a six-item scale (Cronbach's alpha = 0.97) developed by Scott and Bruce (1994). A sample item was "he/she searches out new technologies, processes, techniques and/or product ideas", (from 1 = strongly disagree to 7 = strongly agree).

3.2.6 Control variables. Extant studies suggest that boundary-spanning or innovative behaviour may be related to demographic characteristics (Marrone *et al.*, 2007). De Jonge *et al.* (2012) found that men were likelier to engage in innovative behaviour and that age was significantly positively related to innovative behaviour. Scott and Bruce (1994) found that education was significantly positively related to innovative behaviour and that tenure was significantly negatively related to innovative behaviour. Soni and Bakhru (2019) argue that marriage impacts innovative behaviour through well-being. Therefore, we controlled for gender (1 = female, 2 = male), marital status (1 = single, 2 = married), age (in years), education (1 = middle school education or below, 2 = high school education, 3 = associate's degree, 4 = bachelor's degree, 5 = postgraduate degree) and tenure (in years). We also controlled for team size (number of employees in a team) and team type (1 = finance, 2 = marketing, 3 = manufacturing, 4 = operation, 5 = design, 6 = management, 7 = others) because larger teams have more resources helpful to innovative behaviour and team type is associated with innovative behaviour (Demircioglu, 2020).

3.3 Data analysis

The data were analysed using SPSS (version 22.0; SPSS Inc., Chicago, IL, USA), Mplus 7.4 (Muthén and Muthén, 2012) and R 3.6.0 (<http://www.R-project.org>). First, we performed multilevel confirmatory factor analyses (MCFAs) and computed the average variance extracted (AVE) of the key variables to determine their validity. Second, multilevel structural equation modelling (MSEM) was used to examine the mediating effect of ego depletion (H1). A latent moderated structural equation (LMS) was used to test the moderating effect of intrinsic motivation (H2a), and random coefficient modelling was used to test the moderating effect of organisational support (H3a) (Kelava *et al.*, 2011; Preacher *et al.*, 2016). Third, the moderated mediation process (H2b and H3b) was tested through Monte Carlo simulations using the open-source software R.

4. Results

4.1 Discriminant validity and convergence validity

As organisational support was at the team level, but four other variables were at the individual level, we conducted a standard confirmatory factor analysis (CFA) for individual-level constructs (Model 1 see Table 1) and a standard CFA for organisational support at the team level (Model 2 see Table 1) and then conducted MCFAs using Mplus 7.4. We report a summary of the CFA for Models 1 and 2 in Table 2. According to our findings, the standardised coefficients of factor loading for all items were larger than 0.5. Additionally, the multilevel five-factor model (Model 3, see Table 1) fit the data well. The AVEs for boundary-spanning behaviour, ego depletion, intrinsic motivation, organisational support and innovative behaviour were 0.57, 0.62, 0.88, 0.50 and 0.84, respectively, indicating good convergence validity (Anderson and Gerbing, 1988; Hair et al., 2006). They were larger than the square of all the variables' correlation coefficients (see Table 3), supporting discriminant validity.

4.2 Common method variance

Following Podsakoff et al. (2003), we used a multi-wave and multi-source research design and used a rating scale for intrinsic motivation and ego depletion differing from that of other variables to diminish common method biases.

To assess common method biases, we examined and controlled for an unmeasured latent method factor model at the individual level, with all the items loaded on this latent method factor and the trait factors they were assumed to measure (Podsakoff et al., 2003). A comparison of the unmeasured latent method factor model ($\chi^2 = 554.83$, $df = 141$, $\chi^2/df = 3.93$, RMSEA = 0.08, CFI = 0.96, TLI = 0.95) and Model 1 in Table 1 ($\chi^2 = 684.25$, $df = 146$, $\chi^2/df = 4.69$, RMSEA = 0.08, CFI = 0.95, TLI = 0.94) indicates a significant change in the chi-square value ($\Delta\chi^2(5) = 129.42$, $p < 0.001$). Previous researchers have pointed out that chi-square values are easily affected by the sample size, especially when the sample size is larger than 200 (Cheung and Rensvold, 2002). Accordingly, Little (1997) suggests examining TLI for model choice. If the TLI change is smaller than 0.05, it indicates that adding the unmeasured latent method factor does not significantly improve Model 1. Given that the sample size of the current study was 519, we followed Little's (1997) advice and found that TLI increased by 0.01 when the unmeasured latent method factor was included. This result shows that adding an unmeasured latent method factor did not significantly improve Model 1. The latent variables variance, the method factor variance and the measurement errors variance were 79.06%, 11.47%, 9.47% respectively, thus, common method variance did not have a significant impact on variable relationships.

Table 1.
The results of
multilevel
confirmatory factor
analyses

Model	χ^2	df	χ^2/df	RMSEA	CFI	TLI	SRMR	
							Within	Between
Model 1 (a standard CFA for boundary-spanning behaviour, ego depletion, intrinsic motivation, and innovative behaviour at individual level)	684.25	146	4.69	0.08	0.95	0.94	0.07	
Model 2 (a standard CFA for organisational support at the team level)	16.40	5	3.28	0.07	0.99	0.98	0.02	
Model 3 (hypothesised multilevel five-factor model)	553.78	151	3.67	0.07	0.93	0.92	0.07	0.05
Note(s): Individual level, $N = 519$; team level, $N = 108$								
Source(s): Author's own creation/work								

Items	Standardised coefficients of factor loading	<i>P</i> - values
Model 1		
<i>Boundary-spanning behaviour</i>		
I solicit information and resources from elsewhere in and/or beyond the division	0.52	0.000
I try to influence important actors elsewhere in and/or beyond the division on behalf of the team and its work	0.93	0.000
I make use of their relationships with others on behalf of the team	0.93	0.000
I solicit information and resources beyond what comes through official channels	0.54	0.000
<i>Intrinsic motivation</i>		
Because I enjoy the work itself	0.88	0.000
Because it's fun	0.96	0.000
Because I find the work engaging	0.96	0.000
Because I enjoy it	0.96	0.000
<i>Ego depletion</i>		
My mind feels unfocussed	0.78	0.000
It would take a lot of effort for me to concentrate on something	0.88	0.000
I feel mentally exhausted	0.82	0.000
I feel like my willpower is gone	0.72	0.000
My mental energy is running low	0.73	0.000
<i>Innovative behaviour</i>		
He/she searches out new technologies, processes, technique, and/or product ideas	0.91	0.000
He/she generates creative ideas	0.95	0.000
He/she promotes and champions ideas to others	0.89	0.000
He/she investigates and secures funds needed to implement new ideas	0.89	0.000
He/she develops adequate plans and schedules for the implementation of new ideas	0.93	0.000
He/she is innovative	0.93	0.000
Model 2		
<i>Organisational support</i>		
The organisation takes pride in my accomplishments	0.55	0.000
The organisation really cares about my well-being	0.76	0.000
The organisation values my contributions to its well-being	0.84	0.000
The organisation strongly considers my goals and values	0.82	0.000
The organisation is willing to help me if I need a special favour	0.50	0.000
Note(s): Individual level, <i>N</i> = 519; team level, <i>N</i> = 108		
Source(s): Author's own creation/work		

Table 2.
Summary of the results
of confirmatory factor
analyses for Model 1
and Model 2 in [Table 1](#)

4.3 Descriptive statistics

SPSS 22.0 was used to calculate the correlation matrices of the variables, means and standard deviations (see [Table 3](#)). Notably, boundary-spanning behaviour was not significantly correlated with ego depletion, showing that the relationship between them may be conditional. Ego depletion was significantly and negatively correlated with innovative behaviour ($r = -0.18, p < 0.001$), providing initial support for our argument.

4.4 Hypotheses testing

4.4.1 Testing H1. Following [Preacher et al. \(2010\)](#), we ran two-level structural equation models using Mplus 7.4 to test the mediating effect of ego depletion. The results showed that the model fit the data well ($\chi^2 = 921.52, df = 361, \chi^2/df = 2.55, RMSEA = 0.06, TLI = 0.88$,

Table 3.
Means, standard
deviations and
correlations

	Variables	Means	SD	1	2	3	4	5	6	7	8	9
Individual level	1 Gender-employee	1.45	0.50									
	2 Age-employee	29.87	4.34	0.02								
	3 Education-employee	3.74	0.73	0.06	0.05							
	4 Marriage-employee	1.52	0.50	-0.01	0.58***	-0.05						
	5 Tenure-employee	3.46	3.50	-0.01	0.55***	-0.15**	0.38***					
	6 Boundary-spanning behaviour	4.68	1.00	-0.08	0.09*	0.22***	0.11*	0.14**	(0.57)			
Team level	7 Ego depletion	2.20	0.68	-0.04	0.01	0.08	0.01	0.10*	0.003	(0.62)		
	8 Intrinsic motivation	5.29	1.28	-0.03	0.03	-0.08	0.03	-0.01	0.02	-0.17***	(0.88)	
	9 Innovative behaviour	4.33	1.27	0.04	0.10*	-0.19***	0.10*	0.15**	0.17***	-0.18***	0.11*	(0.84)
	1 Gender-leader	1.55	0.50									
	2 Age-leader	33.94	4.58	0.12								
	3 Education-leader	4.01	0.62	0.13	0.10							
	4 Marriage-leader	1.80	0.40	0.42***	0.37***	0.05						
	5 Tenure-leader	5.77	3.89	0.03	0.33**	-0.05	0.35***					
	6 Team size	27.26	81.73	-0.13	0.10	-0.32**	0.12	0.23*				
	7 Team type	3.94	2.18	-0.24*	0.11	-0.10	-0.12	-0.17	-0.11			
8 Organisational support		4.74	0.55	-0.09	-0.10	-0.03	-0.07	-0.13	-0.11	0.10	(0.50)	
Note(s): Individual level, $N = 519$; team level, $N = 108$; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The values in parenthesis are AVE												
Source(s): Author's own creation/work												

CFI = 0.90). Table 4 shows the unstandardised coefficients. The results showed that boundary-spanning behaviour was negatively associated with ego depletion ($\beta = -0.11$, $p < 0.05$), which does not support our argument. Ego depletion is negatively associated with innovative behaviour ($\beta = -0.14$, $p < 0.05$). The indirect effect of boundary-spanning behaviour on innovative behaviour through ego depletion was not significant (coefficient = 0.02, 95% CI = -0.002 to 0.04). Therefore, H1 was not supported.

4.5 Testing H2a

The grand-centring technique was used to test the moderating effect of intrinsic motivation. An LMS was then employed to examine H2a (Kelava *et al.*, 2011; Preacher *et al.*, 2016).

Variables	Mediating effect of ego depletion		Moderating effect of intrinsic Motivation	Moderating effect of organisational Support
	Ego depletion	Innovative behaviour	Ego depletion	Ego depletion
<i>Individual-level variables</i>				
Eedu1	-0.23	0.03	-0.33*	-0.33*
Eedu2	-0.16	-0.19	-0.14	-0.31
Eedu3	-0.06	-0.01	-0.08	-0.08
Eedu4	0.09	-0.002	0.07	0.08
Egender	-0.04	0.07	-0.03	-0.04
Eage	-0.004	0.01	-0.003	-0.01
Emarriage	0.003	0.06	-0.002	0.02
Etenure	0.03*	0.01	0.03*	0.03**
Boundary-spanning behaviour	-0.11*		-0.09	-0.15*
Ego depletion		-0.14*		
Intrinsic motivation			-0.08*	
Intrinsic motivation $\times$ boundary-spanning behaviour			-0.18**	
<i>Team-level variables</i>				
Ledu1	0.24	-1.00*	0.39*	0.30
Ledu2	-0.03	0.51	-0.06	-0.07
Ledu3	0.14	-0.28	0.21	0.16
Lgender	-0.15	0.29	-0.16	-0.15
Lage	0.01	-0.06*	0.01	-0.002
Lmarriage	-0.26	-0.14	-0.20	-0.12
Ltenure	-0.01	0.02	-0.004	-0.004
Team type1	-0.12	0.22	-0.17	-0.13
Team type2	0.32	0.85	0.11	0.07
Team type3	-0.21	1.31***	-0.14	-0.01
Team type4	0.24	1.16*	0.31	0.31
Team type5	0.35	-0.18	0.31	0.27
Team type6	0.15	0.37	0.12	0.17
Team size	0.0001	0.01**	0.0001	0.0001
Organisational support				-0.36
Organisational support $\times$ boundary-spanning behaviour				-0.83*

Note(s): Individual level, $N = 519$; team level, $N = 108$; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Eedu1-Eedu4 and Ledu1-Ledu3 stand for the dummy variables for employees' education and leaders' education. Team type1-Team type6 stand for the dummy variables for Team type

Source(s): Author's own creation/work

Table 4. Summary of the results of two-level structural equation models

The results showed that the effect of the interaction between boundary-spanning behaviour and intrinsic motivation on ego depletion is significant ($\beta = -0.18, p < 0.01$). Therefore, H2a was supported. We used PROCESS in SPSS to perform the Johnson–Neyman approach to study the conditional effect across all potential values of the moderator and identify regions of significance. The results showed that when the value of intrinsic motivation was between -4.29 and -1.52 , boundary-spanning behaviour had a significantly positive effect on ego depletion. When the value of intrinsic motivation is between 0.19 and 1.71 , boundary-spanning behaviour has a significantly negative effect on ego depletion. We plotted the interaction effects by taking the values of one standard deviation above and below the mean (Figure 2). Taken together, the evidence above indicates that high intrinsic motivation reduces ego depletion caused by engaging in boundary-spanning behaviour. However, low intrinsic motivation increases ego depletion caused by engaging in boundary-spanning behaviour.

4.5.1 Testing H3a. Cross-level interaction and random coefficient modelling have been used to test the moderating effect of organisational support (Preacher et al., 2016). The results showed that the effect of the interaction between boundary-spanning behaviour and organisational support on ego depletion was significant ($\beta = -0.83, p < 0.05$). Therefore, H3a was supported. The results of the Johnson–Neyman approach showed that when the value of organisational support is between -1.72 and -0.89 , boundary-spanning behaviour has a significantly positive effect on ego depletion. When the value of organisational support is between -0.04 and 1.43 , boundary-spanning behaviour has a significantly negative effect on ego depletion. We plotted the interaction effects by taking the values of one standard deviation above and below the mean (Figure 3). Taken together, the evidence above indicates that high organisational support reduces ego depletion caused by engaging in boundary-spanning behaviour. However, low organisational support increases ego depletion caused by engaging in boundary-spanning behaviour.

4.5.2 Testing H2b. To test H2b and H3b, following Edwards and Lambert (2007), we applied the bootstrapping-based analytic approach through Monte Carlo simulation procedures using the open-source software R. In our study, intrinsic motivation (H2b) and organisational support (H3b) served as moderators in the first-stage moderated mediation model. When intrinsic motivation was high, the mediating effect of ego depletion between boundary-spanning behaviour and innovative behaviour was positive and significant (coefficient = 0.08 , 95% CI = 0.01 to 0.17). When intrinsic motivation was low, the mediating effect of ego depletion between boundary-spanning behaviour and innovative behaviour was negative and significant (coefficient = -0.12 , 95% CI = -0.27 to -0.01). Furthermore, a significant difference between the mediation effects in these two groups (i.e. intrinsic

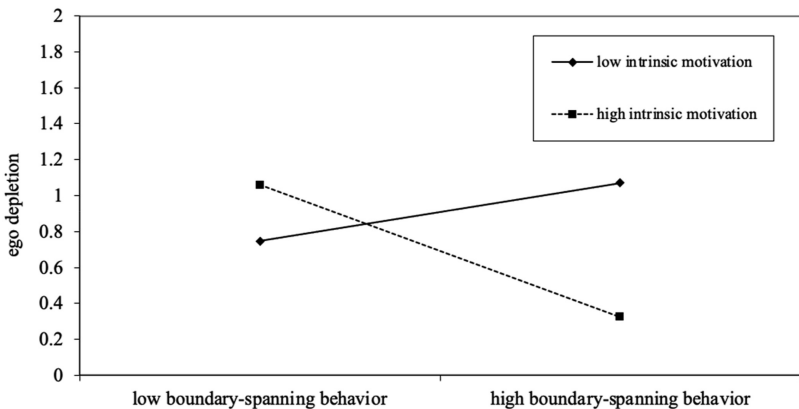
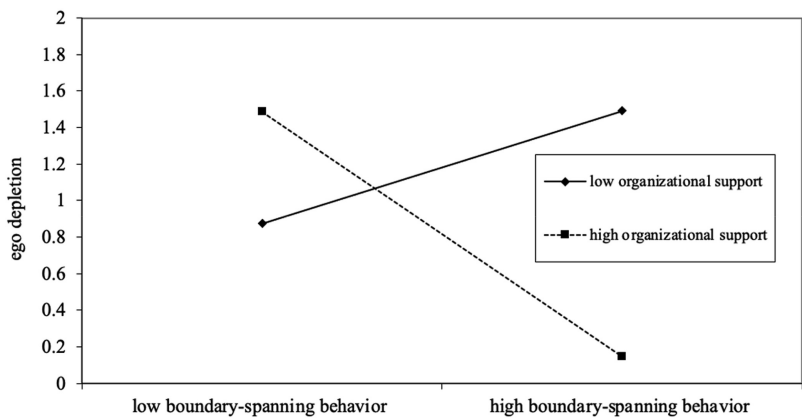


Figure 2.
Interactive effect of
boundary-spanning
behaviour and intrinsic
motivation on ego
depletion

Source(s): Author's own creation/work



Source(s): Author’s own creation/work

Figure 3.
Interactive effect of
boundary-spanning
behaviour and
organizational support
on ego depletion

motivation low vs high) was found (Δ coefficient = 0.20, 95% CI = 0.03 to 0.44), indicating that intrinsic motivation moderated the mediating effect of ego depletion on boundary-spanning behaviour and innovative behaviour. Therefore, H2b was supported.

4.5.3 Testing H3b. When organisational support was high, the mediating effect of ego deletion between boundary-spanning behaviour and innovative behaviour was positive and significant (coefficient = 0.38, 95% CI = 0.01 to 0.94). When organisational support was low, the mediating effect of ego depletion between boundary-spanning behaviour and innovative behaviour was negative and significant (coefficient = -0.40, 90% CI = -0.93 to -0.02). Furthermore, a significant difference between mediation effects in these two groups (i.e. low vs high organisational support) was found (Δ coefficient = 0.79, 90% CI = 0.08 to 1.74), indicating that organisational support moderated the mediating effect of ego depletion between boundary-spanning behaviour and innovative behaviour. Therefore, H3b was supported. We summarise the results of the hypotheses testing (see Table 5) for easier understanding.

Hypotheses	Result
H1: Boundary-spanning behaviour has a negative effect on boundary spanners’ subsequent innovative behaviour via ego depletion	Unsupported
H2a: Intrinsic motivation moderates the relationship between boundary-spanning behaviour and ego depletion, such that the relationship is stronger for boundary spanners with low intrinsic motivation	Supported
H2b: Intrinsic motivation moderates the mediating relationship between boundary-spanning behaviour and innovative behaviour via ego depletion, such that the mediating effect is stronger for boundary spanners with low intrinsic motivation	Supported
H3a: Organisational support moderates the positive relationship between boundary-spanning behaviour and ego depletion, such that the relationship is stronger for boundary spanners with low organisational support	Supported
H3b: Organisational support moderates the mediating relationship between boundary-spanning behaviour and innovative behaviour via ego depletion, such that the mediating effect is stronger for boundary spanners with low organisational support	Supported

Source(s): Author’s own creation/work

Table 5.
The results of testing
hypotheses

5. Discussion

Drawing on ego depletion theory, we examined the dark side of boundary-spanning behaviour on employees' innovative behaviour, as well as its mediating and moderating mechanisms from the perspective of resources. The results showed that when employees' intrinsic motivation or organisational support was low, boundary-spanning behaviour had a significant negative effect on employees' innovative behaviour via ego depletion.

5.1 Theoretical implications

Our research makes several theoretical contributions to boundary-spanning behaviour, innovative behaviour and ego depletion theory. First, the overwhelming majority of findings with existing theory have shown that boundary-spanning behaviour has a positive effect on boundary spanners (Fleming and Waguespack, 2007; Liu *et al.*, 2018; Salem *et al.*, 2018). However, several scholars have recently proposed that boundary-spanning behaviour is likely detrimental to boundary spanners (Marrone *et al.*, 2007; Song *et al.*, 2017). How can we explain the contrasting conclusions regarding the consequences of boundary-spanning behaviour? We argue that intrinsic motivation and organisational support may serve as the underlying boundary conditions. Here, we applied the ego depletion theory to provide a more comprehensive understanding of when boundary-spanning behaviour increases or reduces innovative behaviour. We found that such behaviour is likely to harm individuals' subsequent innovative behaviour when their intrinsic motivation towards such behaviour or organisational support is low. However, boundary-spanning behaviour had a positive effect on subsequent innovative behaviour when intrinsic motivation towards such behaviour or organisational support was high. Our research suggests that the consequence of boundary-spanning behaviour is conditional, featuring a mixture of both positive and negative links between boundary-spanning behaviour and innovative behaviour.

Second, existing studies have reported that organisational climate (Scott and Bruce, 1994), skill (Amabile, 1988), leadership (Carmeli *et al.*, 2013; Chen and Hou, 2016; van Assen and Caniëls, 2022), motivation (Amabile, 1988), individual traits (Raja and Johns, 2010), cognition (Zhou *et al.*, 2009; Tierney and Farmer, 2011; Chen *et al.*, 2022; He and Liu, 2022) and emotions (Fong, 2006) are antecedents of innovative behaviour based on Amabile's (1988) model of creativity and innovation from a static process. However, few studies have explored the impact of previous behaviours on subsequent innovative behaviours. Our research explores how boundary-spanning behaviour impacts subsequent innovative behaviour from a process perspective. Our findings indicate that individual behaviours are not independent of each other but linked together. In other words, individual behaviours are not static but dynamic processes. Therefore, our study enriches the antecedent variables of innovative behaviour from a dynamic perspective.

Third, our research contributes to understanding the specific conditions under which the ego depletion effect is driven. Our work highlights the theoretically and empirically grounded conditions under which the ego depletion effect is especially likely to occur when employees' intrinsic motivation or organisational support is low. These findings extend the boundary conditions of ego depletion theory.

5.2 Practical implications

Our research shows that, under particular conditions, boundary-spanning behaviour is likely to have a positive or negative impact on innovative behaviour. First, this finding offers a clearer lens for leaders to understand the complex boundary conditions under which boundary-spanning behaviour positively or negatively relates to innovative behaviour. Thus, organisations should take measures to help leaders identify the underlying boundary-spanning behaviour to promote the positive effect of boundary-spanning behaviour and

prevent its negative effect. For example, organisations could develop training courses for leaders to teach them useful methods for identifying the underlying boundary conditions of the mixture effects of boundary-spanning behaviour.

Second, our research shows that when organisational support is low, boundary-spanning behaviour has a negative effect on innovative behaviour via ego depletion. Therefore, organisations should take measures to support employees' boundary-spanning behaviour. For example, organisations can support and care for employees as one of the indicators used to assess leader performance. This could encourage leaders to actively help and support employees when they encounter difficulties in performing boundary-spanning behaviour, thereby reducing employees' ego depletion and preventing the adverse effects of boundary-spanning behaviours.

Third, our research shows that when employees' intrinsic motivation towards boundary-spanning behaviour is low, boundary-spanning behaviour has a negative effect on innovative behaviour through ego depletion. Therefore, organisations can recruit new employees with high intrinsic motivation towards boundary-spanning behaviour. The scale for measuring intrinsic motivation can be used to test candidates' intrinsic motivation when organisations recruit new employees.

5.3 Limitations and future research

Our data revealed that performing boundary-spanning behaviour alone, without considering intrinsic motivation and organisational support, does not lead to ego depletion. This implies that other moderations may drive the ego-depletion effect. Future scholars should investigate other boundary conditions. Additionally, H1, which focusses on the mediating effect of ego depletion, was not supported. As a result, future research should continue to examine other mediating mechanisms, such as psychological and emotional mechanisms, that drive the link between boundary-spanning behaviour and innovative or other behaviours.

Notes

1. We also tested all hypotheses without dropping the reverse-scoring item, consistent with the results that dropped the reverse-scoring item. Thus it indicated that the results of our study had high robustness.

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Appendix Scales

Boundary-spanning behaviour

- I solicit information and resources from elsewhere in and/or beyond the division.
- I try to influence important actors elsewhere in and/or beyond the division on behalf of the team and its work.
- I make use of their relationships with others on behalf of the team.
- I solicit information and resources beyond what comes through official channels.

Intrinsic motivation towards boundary-spanning behaviour

- Because I enjoy the work itself.
- Because it's fun.
- Because I find the work engaging.
- Because I enjoy it.

Ego depletion

- My mind feels unfocussed.
- It would take a lot of effort for me to concentrate on something.
- I feel mentally exhausted.
- I feel like my willpower is gone.
- My mental energy is running low.

Organisational support

- The organisation takes pride in my accomplishments.
- The organisation really cares about my well-being.
- The organisation values my contributions to its well-being.
- The organisation strongly considers my goals and values.
- The organisation shows little concern for me (R).
- The organisation is willing to help me if I need a special favour.

Innovative behaviour

- He/she searches out new technologies, processes, technique, and/or product ideas.
- He/she generates creative ideas.
- He/she promotes and champions ideas to others.
- He/she investigates and secures funds needed to implement new ideas.
- He/she develops adequate plans and schedules for the implementation of new ideas.
- He/she is innovative.

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